

1. Administrative & Study Identification

Full Study Title: A Study to Investigate the Effects of Zibotentan/Dapagliflozin Combination Compared to Dapagliflozin Alone in Adult Participants With Chronic Kidney Disease and High Proteinuria **Short Title/Acronym:** ZODIAC **Protocol Number:** AZ-RU-00010 **NCT Number:** NCT06942910 **Sponsor Name:** AstraZeneca **Trial Status:** Recruiting **Investigational Product:** Zibotentan/Dapagliflozin Fixed Dose Combination (FDC) **Phase of Development:** Phase II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the efficacy, safety, and tolerability of zibotentan and dapagliflozin in a Fixed Dose Combination (FDC) compared to dapagliflozin alone, given once daily (QD) on top of Standard of Care (SoC), in adult participants with Chronic Kidney Disease (CKD) and high proteinuria, with or without T2DM. **Secondary Objectives:** To assess the long-term changes in estimated Glomerular Filtration Rate (eGFR), and to closely evaluate safety endpoints, particularly relating to fluid retention and heart failure events.

2.2 Background & Rationale To address the unmet medical need for enhanced renoprotection in CKD patients with high proteinuria. While selective endothelin A (ETA) receptor antagonists have demonstrated benefits in reducing albuminuria, they carry known risks of fluid retention. This study aims to determine if combining the ETA antagonist zibotentan with the SGLT2 inhibitor dapagliflozin can maximize renal function preservation while mitigating the risks of fluid overload and heart failure commonly associated with ERA monotherapy. Furthermore, the results of this study will provide clinical data on efficacy and safety of an innovative treatment in a new region (the Russian Federation), which will be an important additional data source for the Zibotentan/Dapagliflozin FDC approval process in the Eurasian Economic Union.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (Dapagliflozin alone) **Blinding:** Double-blind **Randomization:** 2-arm parallel group, Randomized

3.2 Planned Enrollment & Duration Target **N**: 224 evaluable patients **Number of Sites**: Multicenter (Global, including sites across US, Europe, Australia, and Asia) **Estimated Study Start**: 07-May-2025 **Estimated Primary Completion**: 31-Jul-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. ≥ 18 years of age at the time of signing the informed consent. 2. Diagnosis of CKD with eGFR ≥ 20 and < 90 mL/min/1.73m² AND UACR > 700 mg/g (> 79 mg/mmol) or UPCR > 1000 mg/g (> 113 mg/mmol). 3. Body mass index (BMI) within the range ≤ 40 kg/m². 4. Female participants must be either not of child-bearing potential or women of childbearing potential (WOCBP) using at least one highly effective birth control method for at least 3 months prior to first dose of study intervention. All WOCBP must have a negative serum pregnancy test result at screening. 5. Receiving RAASI therapy (ACEi or ARB), and for the patient maximum tolerated labelled daily dose, that has been stable for at least 4 weeks.

4.2 Exclusion Criteria 1. NYHA class III or class IV Heart Failure (HF); participants hospitalized for HF and/or who have not been stable on HF therapy during the last 6 months prior to screening. 2. HF due to cardiomyopathies that would primarily require other specific treatment, high output HF (e.g., due to hyperthyroidism or Paget's disease), or HF due to primary cardiac valvular disease/dysfunction, severe functional mitral or tricuspid valve insufficiency, or planned cardiac valve repair/replacement. 3. Evidence of rales or jugular venous distention on physical examination. 4. Type 1 diabetes mellitus or clinically significant, unstable, or uncontrolled medical condition. 5. Known hypersensitivity to dapagliflozin, zibotentan, any excipients, SGLT2i therapy, or ERAs. 6. History of any life-threatening ventricular dysrhythmia (continuous or paroxysmal). 7. Participants hospitalised for heart disease or cardiac procedures or for COVID-19 during the last 3 months prior to screening. 8. History of solid organ transplantation or bone marrow transplant; renal replacement therapy or previous kidney transplant. 9. Any condition with a life expectancy of less than 1 year based on investigator's clinical judgment. 10. Malignancy within the past 5 years (excluding non-melanoma skin cancer and curatively treated cervical carcinoma in situ). 11. Significant liver disease or significant impairment of liver function defined as AST or ALT $> 3 \times$ upper limit of normal (ULN) or Total serum bilirubin $> 2 \times$ ULN. 12. Known history of significant drug or alcohol abuse within 12 months of screening. 13. On treatment with strong or moderate CYP3A4 inducer; on systemic immunosuppression therapy other than prespecified stable maintenance therapy. 14. Participants treated or expecting to be treated with tolvaptan, any other ERAs, or budesonide. 15. Systolic blood pressure above 160 mmHg and/or below 90 mmHg. 16. NT-proBNP ≥ 600 pg/mL (or NT-proBNP ≥ 1200 pg/mL, if

associated with atrial fibrillation) measured by local laboratory at screening. 17. Echocardiography at screening showing left ventricular ejection fraction (LVEF) $< 50\%$, significant ventricular wall motion abnormality, or isolated pulmonary arterial hypertension or right ventricular failure. 18. Women who are pregnant, breast-feeding, intending to get pregnant, or unwilling/unable to use adequate contraception.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Zibotentan/Dapagliflozin Fixed Dose Combination (FDC) administered Once Daily (QD) based on eGFR values. **Route of Administration:** Oral **Duration of Treatment:** 28-day run-in with Dapagliflozin (if not already on SGLT2i), followed by a 12-week double-blind period, concluding with a 4-week open-label dapagliflozin phase.

5.2 Concomitant & Prohibited Therapy Permitted: Stable Standard of Care (SoC) regimens for CKD, hypertension, and Type 2 Diabetes Mellitus. **Prohibited:** Strong or moderate CYP3A4 inducers, systemic immunosuppression therapy (other than prespecified stable maintenance therapy), tolvaptan, other ERAs, or budesonide.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, eGFR/UACR Labs, SGLT2i Run-in Period Initiation (if applicable)
Baseline	Day 0	Randomization, First Dose of double-blind FDC or Active Comparator
Interim	Weeks 2 to 12	Safety Labs, Efficacy Assessment (UACR, eGFR), AE/Fluid Retention Monitoring
End of Study	Week 16	Final Vital Signs, Exit Interview, IP Accountability, Open-Label Transition

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in log-transformed Urinary Albumin to Creatinine Ratio (UACR) from baseline, estimating the efficacy of zibotentan and dapagliflozin in FDC versus dapagliflozin alone in reducing albuminuria. **Measurement Method:** Central Laboratory Analysis of urine collections.

7.2 Secondary & Exploratory Endpoints

1. Change in log-transformed Urinary Protein to Creatinine Ratio (UPCR) from baseline.
2. Change in systolic and diastolic blood pressure (BP) from baseline.
3. Number of participants experiencing adverse events.
4. Change from baseline in estimated Glomerular Filtration Rate (eGFR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via HCP assessment, laboratory abnormalities, and patient reporting (Number of participants experiencing adverse events), with special attention to fluid overload and heart failure symptoms. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (DMC) will monitor the safety and scientific integrity of the trial to ensure patient wellbeing.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy measures; Safety Set for all adverse event analyses. **Sample Size Justification:** $N=224$ is appropriately powered ($\alpha = 0.05$, $\text{Power} = \geq 80\%$) to detect a statistically significant reduction in UACR between the zibotentan/dapagliflozin combination and dapagliflozin alone. **Handling of Missing Data:** Mixed-Model for Repeated Measures (MMRM) and/or Multiple Imputation for continuous endpoints such as UACR and eGFR.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence, drug accountability, and patient safety. **Audit Trail:** Robust electronic Case Report Form (eCRF) locking procedures, source data verification (SDV), and data clarification forms.

11. Study Identifiers & Governance

Primary ID: AZ-RU-00010 **Registry Number:** NCT06942910 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ZODIAC **Official Regulatory Title:** A Multicentre, Randomised, Double-blind, Active-Controlled, 2-arm Parallel-group Treatment, Phase II Study to Evaluate the Efficacy, Safety, and Tolerability of Zibotentan/Dapagliflozin Compared to Dapagliflozin Alone in Adult Participants With Chronic Kidney Disease and High Proteinuria

12. Clinical Framework (CKD & Proteinuria Specific)

Primary Condition: Chronic Kidney Disease (CKD) with High Proteinuria **Diagnosis Threshold:** eGFR ≥ 20 and < 90 mL/min/1.73m²; UACR > 700 mg/g **Medical Methodology:** Guideline

Determined Medical Therapy (GDMT) including maximum tolerated RASi **Optimization Target:** Significant UACR reduction and kidney function (eGFR) preservation

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for CKD utilizing combination ERA and SGLT2i therapy **Education Protocol (HCP):** Protocol training on identifying and managing treatment-emergent fluid retention **Education Protocol (Patient):** Education on study drug compliance, dietary sodium moderation, and signs of fluid overload **Quality Audit Mechanism:** Periodic central data reviews and site-level monitoring

14. Observation & Follow-Up Schedule

Total Duration: 16 weeks (12-week double-blind + 4-week open-label) **Onsite Visit Cadence:** Baseline, and routine visits throughout the 16-week period **Remote Monitoring Frequency:** Regular interval check-ins (e.g., Q4W) for ongoing safety surveillance **Checklist Review Interval:** Review of concomitant medications and adverse events at every clinical interaction

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				